

Connor B. Savugot

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Aegis Power Systems, Inc.

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Dear Mrs. Vaughn,

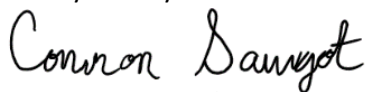
It is my understanding that your company hosts summer interns from NC State. After studying your website, I feel I may be a perfect candidate for an internship with your amazing company. I am currently a sophomore at NC State and am studying computer engineering. As you may know, this degree affords me the opportunity to study a variety of programming languages, electrical engineering, and computer hardware. I am excited about the possibility of learning more about power systems, not only through school, but also through spending the summer working with a company like yours. I feel hands-on learning helps bring understanding to difficult concepts. I am meticulous and hard working. I love to learn and am always excited to expand my current knowledge base and prefer learning in real-world situations. Not only would this be the perfect learning opportunity, but Aegis is located about 15 minutes from my home.

Circuit analysis classes have taught me to analytically solve circuits using Ohm's law and Kirchoff's laws. I have learned to build half-wave and full-wave rectifier circuits. My understanding and use of components used in AC to DC conversion has grown throughout the semester. My comprehension of diodes, capacitors, inductors, and Op amps has grown exponentially this year. Utilizing VRMS to determine the power supply provision and usage within a circuit has been a practiced concept this semester as well. At this time, my understanding of 3-phase power systems is still a little weak, but as the semester progresses, I feel my understanding will grow as it is part of our curriculum planned for later this semester. I have been studying logic circuits for about the past year. Through my classes, I have learned to use logical axioms to manipulate logical expressions that represent different logical circuits, and how to make circuits more efficient by lowering the number of logic gates using Karnaugh maps. I am learning how to represent input and output in solving real world problems.

I am extremely passionate about programming and learning more about any device that runs on 1's & 0's. Currently, I am learning another two programming languages, C++ and Verilog. I also have experience programming in Visual Basic Advanced, Python, MATLAB, AHK, Ti-Basic, LC3 Assembly, and C. I have always enjoyed problem solving and being challenged with puzzles to solve. Growing up in a rural area has helped push me to further develop my online research and design experimentation skills. Often, I had to figure things out on my own, because there was no where I could go for help or advice.

I have 2 summers of experience as an engineering intern at Team Industries. Team Industries designs and produces CNC machined gears, pinions, and sprockets for many well-known customers including Polaris, Harley, and Boeing. During my time there, I learned a lot about the product development process. I had to create several spreadsheets using advanced formulas, conditional formatting, and even programming in Visual Basic Advanced. I also learned how to perform monotonous tasks such as inspecting thousands of gears very effectively by experimenting with different work flows to find what worked best for me. These gears had to be inspected because we found a large quantity of them were outside their allowed tolerances. We had to experiment and revise several aspects of our production process to determine the root cause. We found that the oil in one of our three heat-treat chambers, which we use to harden the machined parts, was old. This caused the parts to be inconsistently out of spec based on which chamber was used. This taught me a lot about problem solving in a professional environment.

Thank you for your time and consideration for any internships you have available,



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